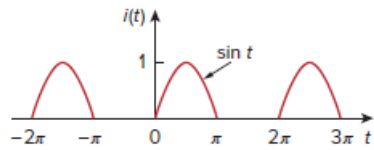


## Problem

Write a computer program to calculate the exponential Fourier series of the half-wave rectified sinusoidal current of Fig. 17.82. Consider terms up to the 10th harmonic.

**Figure 17.82:**



## Step-by-step solution

## Step 1 of 2

$$C_n = \frac{[1 + e^{-jn\pi}]}{[2\pi[1 - n^2]]}$$

This is calculated using the for tram program shown below. The result are also shown.

C for problem 17.30

Complex X, C (0:20)

PIE = 3.1415927

A = 2.0 \* PIE

DO 10 N = 0.10

IR (4-EQ.1) GO TO 10

X = CMPLX(0, PIE \* F10A+ (N))

$$C(N) = \frac{(1.0 + CEXP(-X))}{(A * (1 - F10A(N * N)))}$$

PRINT \*, M, C (N)

10 CONTINUE

STOP

END

| N | Cn                            |
|---|-------------------------------|
| 0 | 0.3188+j0                     |
| 1 | 0                             |
| 2 | -0.1061+j0                    |
| 3 | 0                             |
| 4 | $-0.2121 \times 10^{-1} + j0$ |

## Step 2 of 2

|    |                               |
|----|-------------------------------|
| 5  | 0                             |
| 6  | $-0.9095 \times 10^{-2} + j0$ |
| 7  | 0                             |
| 8  | $-0.5052 \times 10^{-2} + j0$ |
| 9  | 0                             |
| 10 | $-0.3215 \times 10^{-2} + j0$ |